# Research Proposal

Biomedical Data Design Rank Order Assignment

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September 3, 2026

## 1 Project Overview

Given a set of $n$ doctors indexed on D $= \{1, \ldots, n\}$ with a set of $m$ hospitals indexed on H $= \{1, \ldots, m\}$ where $n \geq m$. Each hospital has capacity C: $H \to \mathbb{Z}$. Each doctor $D_i$ has preferences of hospitals written as a permutation $\pi_i \to H$ where $\pi_i(h)$ denotes the rank a doctor $D_i$ assigns to hospital h. We aim to produce assignment function $f : D \to H$ such that each doctor is assigned to a hospital considering their orderings.

We aim to minimize cost

$$C(f) = \sum_{i=1}^{n} \pi_i(f(D_i))$$

And choose

$$f^* = \arg\min_f C(f)$$

### 1.1 Assumptions

In doing so we assume the following

1. All individuals are given equal weighting on their preferences
2. Costs are comparable between individuals
3. We assume that the cost of an action is inversely proportional to its rank

## 2 Methods

We will implement the Hungarian Algorithm for rank order assignment [1].

The algorithm involves the following steps:

1. Construct the cost matrix $C$, where

$$C_{ij} = \pi_i(H_j).$$

2. Subtract the minimum value in each row from every entry in that row. Then, subtract the minimum value in each column from every entry in that column.

3. Identify the minimum number of horizontal and vertical lines needed to cover all zero-valued entries in the resulting matrix.

4. If $n$ lines are required to cover all zeros, select $n$ independent zeros (no two in the same row or column). These zeros define the optimal assignment.

5. If fewer than $n$ lines are required, find the smallest uncovered value. Subtract this value from every uncovered entry and add it to every entry covered by two lines. Leave all other entries unchanged.

6. Repeat Steps 3–5 until $n$ independent zeros can be selected. Assign each doctor to the hospital corresponding to their selected zero.

The resulting assignment $f^*$ minimizes the total preference cost:

$$f^* = \arg\min_f \sum_{i=1}^{n} \pi_i(f(D_i)).$$

## 3 Limitations and Extensions

Further considerations/limitations include

- Currently we assume an inversely proportional relationship between rank and cost, this is not necessarily true. Most generally each doctor has a unique cost for each rank.
- We assume there are more doctors than hospitals such that each hospital is assigned at least 1 doctor. This assumption may be challenged further

## 4 References

## References

[1] Kuhn, H. W. (1955). The Hungarian Method for the Assignment Problem.